

launchSettings.json — Complete Detailed Guide

What is `launchSettings.json`?

`launchSettings.json` is a configuration file used in ASP.NET Core during **local development**.

It tells Visual Studio / .NET CLI:

- How to start the application
- Which URL and port to use
- Whether to use HTTP or HTTPS
- Which environment to run in
- Whether browser should auto-open
- Whether app runs using IIS Express or Kestrel

File Location

Properties/launchSettings.json

Important Understanding

This file is:

- ✓ Used only during development
- ✓ Used locally
- ✓ Not used in production deployment

Real Purpose

When you press:

F5

or

```
dotnet run
```

this file controls:

- App startup behavior
- Ports
- Browser launch
- Environment variables

Example launchSettings.json

```
{
  "$schema": "http://json.schemastore.org/launchsettings.json",
  "iisSettings": {
    "windowsAuthentication": false,
    "anonymousAuthentication": true,
    "iisExpress": {
      "applicationUrl": "http://localhost:36360",
      "sslPort": 44313
    }
  },
  "profiles": {
    "http": {
      "commandName": "Project",
      "dotnetRunMessages": true,
      "launchBrowser": true,
      "launchUrl": "weatherforecast",
      "applicationUrl": "http://localhost:5217",
      "environmentVariables": {
        "ASPNETCORE_ENVIRONMENT": "Development"
      }
    },
    "https": {
      "commandName": "Project",
      "dotnetRunMessages": true,
      "launchBrowser": true,
      "launchUrl": "weatherforecast",
      "applicationUrl": "https://localhost:7123;http://localhost:5217",
      "environmentVariables": {
        "ASPNETCORE_ENVIRONMENT": "Development"
      }
    },
    "IIS Express": {
```

```
    "commandName": "IISExpress",
    "launchBrowser": true,
    "launchUrl": "weatherforecast",
    "environmentVariables": {
      "ASPNETCORE_ENVIRONMENT": "Development"
    }
  }
}
```

1. \$schema

```
"$schema": "http://json.schemastore.org/launchsettings.json"
```

Purpose

Provides JSON schema support.

What it does?

Helps IDE understand:

- Valid properties
- Auto suggestions
- IntelliSense
- Validation

Without this?

Application still works.

But:

- No IntelliSense
- No validation help

2. `iisSettings`

```
"iisSettings": {
```

Purpose

Contains IIS Express configuration.

Used when app runs through:

```
IIS Express
```

instead of Kestrel directly.

3. `windowsAuthentication`

```
"windowsAuthentication": false
```

Purpose

Controls Windows Authentication.

If true

User login uses Windows credentials.

Example:

- Corporate intranet apps
 - Active Directory environments
-

If false

Windows login disabled.

4. anonymousAuthentication

```
"anonymousAuthentication": true
```

Purpose

Allows users without login.

If true

Anyone can access app.

If false

Must authenticate first.

5. iisExpress

```
"iisExpress": {
```

Purpose

Configuration specific to IIS Express.

6. applicationUrl

```
"applicationUrl": "http://localhost:36360"
```

Purpose

Defines application URL.

Meaning

App runs at:

```
http://localhost:36360
```

7. sslPort

```
"sslPort": 44313
```

Purpose

HTTPS port for IIS Express.

Meaning

Secure app URL:

```
https://localhost:44313
```

8. profiles

```
"profiles": {
```

MOST IMPORTANT SECTION

Profiles define:

Different ways to run the application.

Your Profiles

Profile	Usage
http	HTTP mode
https	HTTPS mode
IIS Express	Run through IIS Express

9. http Profile

```
"http": {
```

Purpose

Runs application using plain HTTP.

10. commandName

```
"commandName": "Project"
```

Meaning

Run using built-in Kestrel server.

Other Possible Values

Value	Meaning
Project	Run using Kestrel
IISExpress	Run using IIS Express

11. `dotnetRunMessages`

```
"dotnetRunMessages": true
```

Purpose

Shows startup messages in console.

Example

```
Now listening on: http://localhost:5217
```

12. `launchBrowser`

```
"launchBrowser": true
```

Purpose

Automatically opens browser when app starts.

If true

Browser opens automatically.

If false

Must manually open URL.

13. `launchUrl`

```
"launchUrl": "weatherforecast"
```


Purpose

Browser opens this endpoint automatically.

Final URL

```
http://localhost:5217/weatherforecast
```

14. applicationUrl

```
"applicationUrl": "http://localhost:5217"
```

Purpose

Defines app listening URL.

Meaning

Kestrel server listens here.

15. environmentVariables

```
"environmentVariables": {
```

Purpose

Defines environment variables.

16. ASPNETCORE_ENVIRONMENT

```
"ASPNETCORE_ENVIRONMENT": "Development"
```

VERY IMPORTANT

Controls application environment.

Possible Values

Environment	Purpose
Development	Local development
Staging	Testing server
Production	Live application

Development Environment

Enables:

- Detailed errors
 - Swagger
 - Debugging
 - Developer exception page
-

Production Environment

Disables:

- Detailed exception messages
- Debug info

Improves:

- Security
 - Performance
-

Example Usage in Program.cs

```
if (app.Environment.IsDevelopment())
{
    app.UseSwagger();
}
```

HTTPS Profile

```
"https": {
```

Difference

Supports both:

```
https://localhost:7123
http://localhost:5217
```

IIS Express Profile

```
"IIS Express": {
```

Purpose

Runs app using IIS Express instead of Kestrel.

Difference Between Kestrel vs IIS Express

Kestrel	IIS Express
Cross-platform	Windows only
Lightweight	IIS integrated

Kestrel	IIS Express
Used in production	Mostly development
Faster startup	More IIS features

Internal Flow

When you press:

F5

Visual Studio:

```
Reads launchSettings.json
  ↓
Chooses selected profile
  ↓
Starts server
  ↓
Sets environment variables
  ↓
Launches browser
```

Important Notes

This file is NOT deployed

Production server ignores this file.

Production Uses

Production uses:

- IIS configuration
- Azure settings
- Docker ENV variables
- Linux environment variables

Common Real-World Usage

Developers use this file for:

- Different ports
 - HTTPS setup
 - Debug profiles
 - Multiple environments
 - Swagger startup
-

Most Important Understanding

Section	Purpose
iisSettings	IIS Express config
profiles	Startup configurations
applicationUrl	Server URL
launchBrowser	Auto-open browser
environmentVariables	App environment
ASPNETCORE_ENVIRONMENT	Dev/Prod behavior

Senior Developer Notes

launchSettings.json is:

- ✓ Development-only configuration
 - ✓ Local machine startup settings
 - ✓ Not production configuration
-

Real Interview Answer

What is launchSettings.json?

launchSettings.json is a development-time configuration file in ASP.NET Core used to define how the application launches locally, including URLs, ports, environments, authentication settings, and startup profiles.

Final Summary

`launchSettings.json` controls:

- How app starts
- Which server runs
- Which port used
- Which environment active
- Browser launch behavior

It acts like:

"Startup Configuration Blueprint" for Local Development.